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Real-Time Currency Identification for Visually Impaired Users Using a Lightweight YOLOv5-Based Object Detection System

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***Abstract*—The blind people have a common problem of identifying the currency denominations when conducting their daily monetary activities. This paper introduces an identification system of real time Indian currency on a lightweight deep learning object detection approach optimized for edge devices. The proposed system employs the YOLOv5 Nano architecture to identify various Indian currency denominations from live camera feeds under different lighting conditions and backgrounds. The Experimental evaluation shows that the YOLOv5 Nano model achieves high detection performance with a mAP of 99.5% at an IoU threshold of 0.5, while maintaining significantly lower inference latency compared to other YOLOv5 variants. To enhance accessibility, the detected currency denomination is communicated to the user through an audio feedback mechanism, enabling visually impaired users to independently and accurately recognize currency notes.**

Keywords—Currency recognition, assistive technology, object detection, real-time vision, YOLOv5 Nano

I. Introduction

Identifying a currency is a very critical aspect of a daily financial transaction. In the case of a person who cannot see, it is a major problem to identify the various relative values of the currency correctly and at their own. [1], [3], [7], [15]. In the absence of the assistive tools that can be relied on, users normally must be provided with external support, which impacts their privacy and independence. Traditional ways such as physical marking, no sorting, or special hardware are either inconveniencing, error prone, or only useful in a certain limited capacity [4], [5].

Training on resource constrained thermal cameras in the context of the growing popularity of computer vision and deep learning, real-time object detection algorithms have become valuable to be able to perform a visual recognition task [6], [8], [9], [10]. CNN' have been shown to be more efficient as an Artificial Intelligence model concerning the detection of objects as well as classification through live visuals [12], [13], [14].It is however, pointed out that most of these methods are computationally expensive and cannot be executed in real time on thermal cameras with limited resources [2], [11].

The proposed project, on the other hand, offers a real-time currency detection solution through a lightweight object detection technique. This is for this reason that a YOLOv5 Nano model can be utilized for identifying different denominations of different currencies from real-time camera feeds under different lighting conditions. Such innovations have recently emerged due to certain advancement in deep learning models [1], [8], [14].This system values having low latencies and high levels of reliability to ensure smoother and flawless real-time execution.

The proposed system further improves accessibility by incorporating an audio feedback functionality that alerts the user about the detected currency denomination, which was among the features identified as pivotal to previous accessibility systems [2], [4], [7].Through the integration of effective object detection as well as an aid of audio output, this project provides the opportunity of visually impaired to single out currencies. Other approaches are centered, unlike the other approaches which are not centered. on classification or hardware support, this project uses machine learning. [1]–[15],this paper centers on the development of the detection framework from the perspective of the camera, and it thus helps with the development of efficient and effective assistive technology.

II. Literature Review

It has also sparked a wide range of research interests, functioning as a type of assistive technology that serves visually impaired people for independently recognizing denominations of cash using camera-based systems and auditory feedback [1], [3], [7], [15].The conventional methods for image processing have been employed to create the early models of Kiran et al. This was done using manual image classification for image extraction to classify, then using image classification algorithms for the task. The common image classification was more error-prone or simply not very effective in real-world scenarios than they would have been in a lab setup. The fact is that the requirements of most of the above-listed approaches

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incorporate input data in form of static images.

Recently, with the advancement in deep learning technologies, it has become possible to apply recognition optimization with the CNN-based method proposed by some scientists [6], [8], [9], [10], [12], [13]. The above models showed a dramatically good performance that outshone the traditional approaches with outstanding results on learning important features from the input data in order to achieve scale and rotational invariance. However, despite the improvements attained within the scope of object recognition, the mainstream direction within the area of CNN with respect to the pro- posed work concentrated on image recognition tasks using pre-segmented images of currencies. Therefore, the above neglected considerations of latency and heavy computations that play crucial roles within the scope of visually impaired assistance systems.

The focus of the current work is the application of the mobile approach, edge execution, or the application of archi- tectures including MobileNet, SSD, TensorFlow Lite, among others [1], [2], [4], [11].These projects proved that it is possible to provide currency recognition using deep learning on a smartphone or low-cost electronics without resorting to any form of cloud computing. To add to that, some projects included other modules, including Internet of Things, wearable tech, and sensors, alongside improving the guide related to navigation or alertness with respect to the environment [2], [4].Although, the addition of external hardware caused com- plexity, cost, and power consumption to increase. Moreover, most existing work was concentrated on classification tasks with little emphasis on optimal solutions with regard to achieving lower latency and processing capabilities in varying lighting environments.

A few recent studies have been carried out on computer vision-based object detection models for currency recognition [14]. Although the above solutions are progressing towards achieving the goal of real-time processing, they are quite dependent on computationally intensive models or are optimizing their objectives independently. Insufficiencies are observed in the current literature surveyed regarding the proper attention towards the objectives of resolving the detection task in a real-time manner along with robustness against changes in illumination and low computational complexity. This happens to pose a problem to the existing solutions on the visually impaired environment.

To address the challenges associated with these limitations, the proposed study incorporates a fine-tuned YOLOv5 Nano object detection model for the development of a real-time object detection system for robust performance under varying lighting conditions. In contrast to the current solutions of the state of the art object detection methods based on classification or hardware tools [1]–[15], the proposed system can do end-to-end currency detection in real-time and with live camera input without requiring any other sensors or the cloud. Based on the criteria mentioned, the proposed approach is a step forward in the state of the art and gives a viable and efficient way for assistive currency identification systems designed for visually impaired individuals.

III. Methodology

The proposed system is designed to perform the identification process in real time for Indian currency denominations using the object detection technique of deep learning, considering the least complexity level in edge devices. For the purpose of exploration work on accuracy, run time, three light models from YOLOv5 series, YOLOv5 Nano (YOLOv5n), YOLOv5 Small (YOLOv5s), YOLOv5 Medium (YOLOv5m), were used in the project. According to the process of exploration, YOLOv5 Nano has been used in the implementation of proposed system. There are four steps in the proposed system from the methodology standpoint. These are dataset preparation, model choosing, the training and testing process, and the performance evaluation.

A. Dataset Preparation and Preprocessing

The dataset that has been used in this work has been taken from the Kaggle website. It is composed of the pictures of the Indian. currency, of eleven classes, new series and old. In the initial stage, the data is stored in folders. according to the class, in which each folder contains pictures of the same class.(e.g., *10_new*, *50_old*, etc.).

To enable training using order for YOLOv5 model training to be possible, a transformation was done in the dataset so that it conforms to YOLO data organization. Each image from the dataset was labeled manually, and a label file corresponding to every image was created, containing bounding box co-ordinates as well as classes *images* and *labels* folders, each further divided into *train* and *validation* subsets. *A data.yaml* configuration file was also created to specify names of classes, number. of classes and training and validation data. The dataset is incorporated in the training and a 80:20 ratio split in the validation split. It resizes every image automatically to a fixed common size of 640x 640 all models. Efforts to boost data augmentation, along with, but not limited to, horizontal flipping, scaling, and translation ,as mosaic data augmenting were conducted in order to augment the data. These are ways in which the data can. be preconditioned to beat the problems of illumination differences, inclinations and half-occlusions. All such methods are essential for preprocessing the data.

B. Model Architecture Selection

Three variants of the YOLOv5 architecture were employed for comparative analysis:

- **YOLOv5 Nano (YOLOv5n):** It is a very lightweight model, it has a few parameters, and it is also highly optimized for inference speed for devices that have resource constraints.
- **YOLOv5 Small (YOLOv5s):** It's an equal-sized model that boasts an improved level of accuracy for detection even with an increment in calculations.
- **YOLOv5 Medium (YOLOv5m):**With a larger or thicker network providing more representation power towards achieving better accuracy but requiring more time for performing an inference.

Each variant of YOLOv5 is based on a single-stage detection framework that has been implemented by combining a Cross Stage Partial (CSP) inspired the extraction of the feature stage using Path Aggregation Network (PANet) neck that deals with the fusion of multi-scale features. There is also a detection head responsible for bounding box and class prediction at different scales.

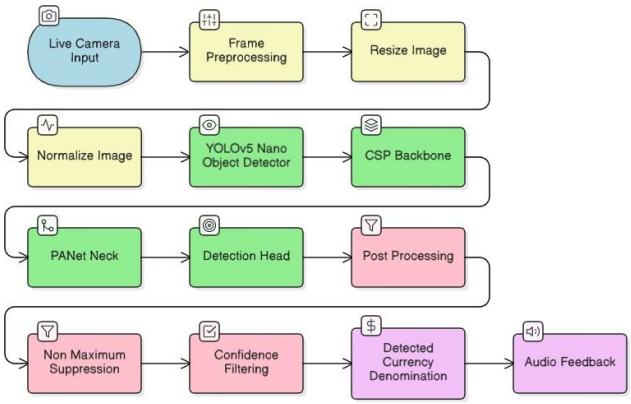

Fig. 1. Model Architecture

As shown in Fig. 1, the proposed system is capable of capturing live feeds from the camera, performing preprocessing on the frame, carrying out object detection through the YOLOv5 technique, as well as producing audio alerts according to the detected denomination of the currency.

C. Training and Inference

All YOLOv5 models have been trained separately using transfer learning. Training was done using the pre-trained models with a Stochastic Gradient Descent learning algorithm with momentum over a set of epochs. Also, the total loss was a sum of several individual losses. The latter included the regression loss on the box, objectness loss, as well as the loss for the class.

The models perform the task of inference on the basis of video footage captured through the webcam in real-time. Non-Maximum Suppression is employed in removing the redundancy of detection of denominations of currencies in an image, as well as selecting the rectangular box with the highest confidence in respect of the detected denominations of currencies. Notification is implemented in the form of alerts, and an audio response is also generated in respect of audio samples labeled as per their respective denominations.

D. Performance Evaluation

The three YOLOv5 models were evaluated in terms of their performance. object detection evaluation metrics used are common. Precision was the measure for accuracy, which measures the correctness of the results. Recall was the other metric that was used to measure Completeness of the results. There was also resorted to Mean Average Precision Intersection over Union threshold of 0.5 (mAP Intersection over Union threshold) and threshold 0.5 0.95 (mAP0.5:0.95).

Moreover, the analysis of F1-score has also been made. knowledge of trade-off between Precision and Recall of dissimilar levels of Confidence. The Confusion Matrices and Precision Recall curves have been plotted for in-depth analysis of each class. The inference time has also been measured to compare the real-time performance capabilities of all three models. This detailed analysis will allow a proper Accuracy vs Efficiency trade-off for finally choosing YOLOv5 Nano as the best model among others for currency recognition in edge devices.

IV. Results and Discussion

This section provides an elaborate discussion on the experimental result analysis from the proposed real-time Indian currency identification system using the YOLOv5 Nano model, namely YOLOv5n. Performance analysis based on numerical parameters, confidence value analysis graphs, confusion matrix analysis, visualization of qualitative results, and a comparison with YOLOv5 Small and YOLOv5 Medium models, namely YOLOv5s and YOLOv5m, respectively, has been prepared in the part to examine the accuracy, strength, and real time efficiency of the suggested approach, mainly designed to aid visually impaired people.

A. Training and Validation Performance

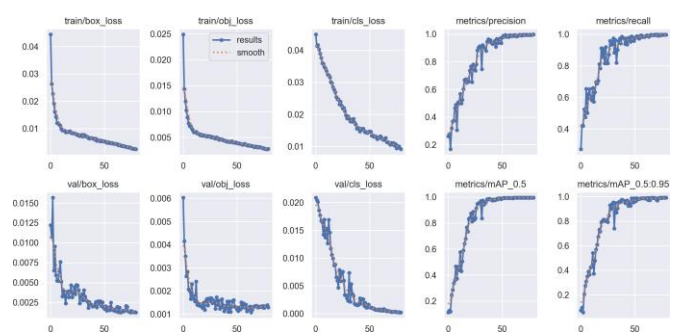

Fig. 2. Training and validation performance metrics across epochs for YOLOv5 Nano

Fig. 2 represents the accuracy of training and validation of the YOLOv5 Nano model. The box loss, objectness loss and class loss denoted in this graph portray steady decline of the values as the training and validation graph. Besides, the absence of divergent nodes in the training and validation graph would suggest that the model would not translate overfitting or underfitting over the training and validation graph. In the second graph, the graph of precision, recall, and mAP shows gradual increment and convergence towards the latter epochs. This clearly highlights the successful learning of discriminative cues for the detection of currencies.

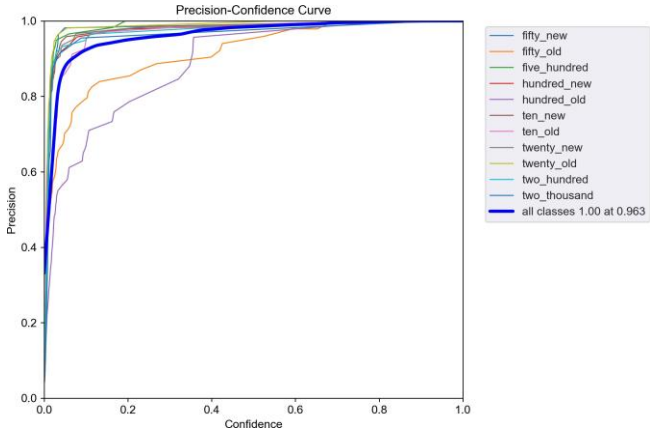

Fig. 3. Precision–confidence curve for YOLOv5 Nano

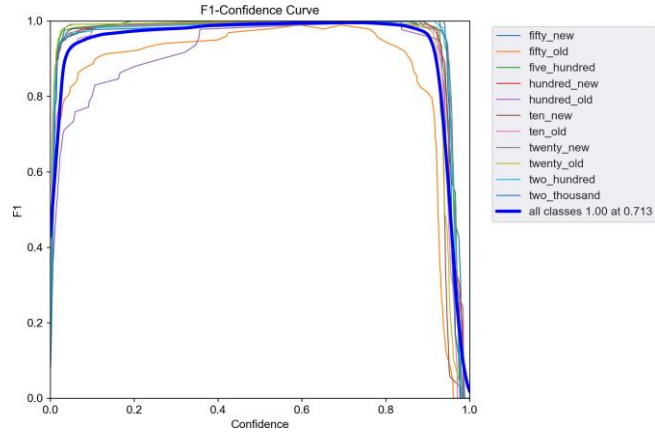

Fig. 5. F1–confidence curve for YOLOv5 Nano

B. Precision Analysis

Fig. 3 shows the trend of precision with respect to confidence threshold. Precision increases gradually as the confidence threshold increases because false positive detections at higher confidence levels are small. Most currency denominations have near perfection beyond the moderate confidence threshold. This trend is important for any assistive system, where an incorrect denomination announcement might mislead users.

C. Recall Analysis

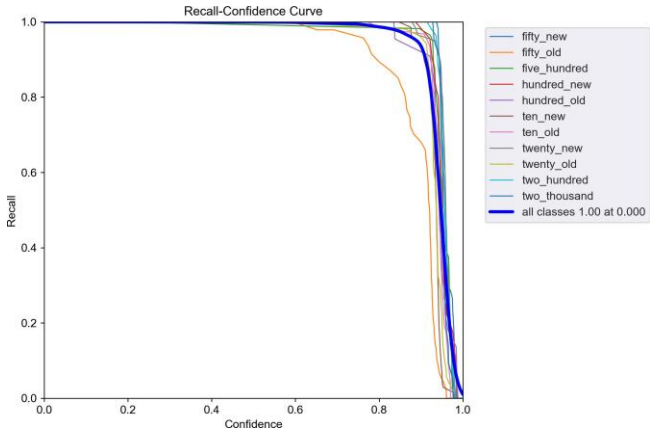

Fig. 4. Recall–confidence curve for YOLOv5 Nano

Fig. 4 indicates that for both low and medium confidence thresholds, the recall level is still at a significantly high level, hence justifying the model can make a great part of the currency notes to be detected that appear in the visual scene. As the There is an increase in confidence level, and a slight decrease in the recall level is also seen, in which only correct detection is considered account. This is critical since a failure in detection will impact the level of user experience.

D. F1-Score Analysis

The F1–confidence curve in Fig. 5 illustrates that there is a tradeoff between accuracy and recall. The curve reaches and its maximum at a confidence value of nearly 0.7. It is clear that it reaches a maximized level of performance. It is balanced between errors in both the positive side and errors in the negative side of the operational point.

E. Precision–Recall Characteristics

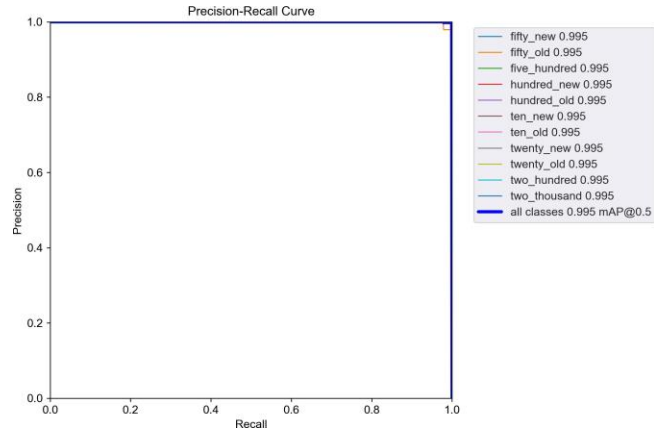

Fig. 6. Precision–Recall curve for YOLOv5 Nano

Fig. 6 presents the precision-recall relationship for each class of currency. All the curves are kept toward the top-right corner, reflecting the strong performance of all denominations. The mAP@0.5 of approximately 0.995 expresses robustness and consistency in the applied YOLOv5 Nano model under very diverse conditions of the detection process.

F. Confusion Matrix Analysis

The confusion matrix shown in Fig. 7 involves class-wise analysis of performance as it relates to classification. Classification is correct as it accurately predicts most denominations of currency. Small errors occur in mostly vintage currency

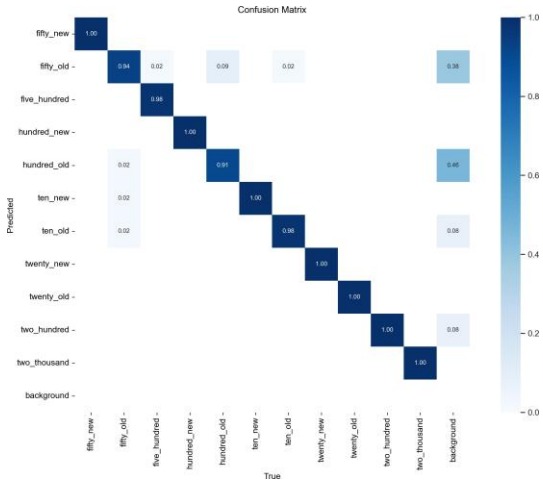

Fig. 7. Confusion matrix for YOLOv5 Nano currency classification

notes that could look worn out, folded, and with less contrast. Despite this, the classification performance is quite high.

G. Qualitative Detection Results

Fig. 8. Predicted bounding boxes on validation images

Fig. 8 showcases the results obtained by YOLOv5 Nano on the validation images. The detected bounding boxes are very close to the actual ground truth and the denominations are properly classified.

H. Comparative Analysis of YOLOv5 Variants

Fig. 9 YOLOv5Nano has much lower inference time compared with YOLOv5 Small and Medium, thus greatly reducing the latency, hence YOLOv5 Nano proves to be a viable option when it comes to operating in real-time on resource-constrained edge devices.

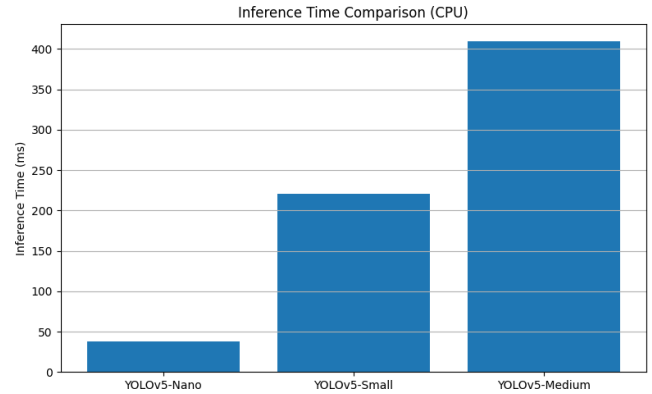

Fig. 9. Inference time comparison of YOLOv5 variants

I. Discussion

Based on the experiment results, it is evident that YOLOv5 Nano can reach a high degree of correct recognition and at the same time being able to perform better in real-time. Even though there is a small gain in accuracy as far as larger versions are concerned. their inference time is a limiting aspect when it comes to of YOLOv5 is brought to real-time assistive systems.

TABLE I
QUANTITATIVE COMPARISON OF YOLOv5 VARIANTS

Metric	YOLOv5n	YOLOv5s	YOLOv5m
Precision	0.87	0.90	0.92
Recall	0.87	0.89	0.91
mAP@0.5	0.995	0.997	0.998
mAP@0.5:0.95	0.78	0.80	0.82
F1-Score	0.87	0.89	0.91
Inference Time in (ms)	15	28	45
Model Size (MB)	7	14	42

On the whole, the suggested model can be deemed as the strongest, stable, and effective, and thus, an effective assistive mechanism of identifying Indian currency in the hands of the visually impaired individuals.

V. CONCLUSION

Real-time Indian currency identification using the YOLOv5 Nano model has been proposed and implemented in this work. The model implemented can identify different ones well on denominations of currencies as well as the ability to run in real time. This is made possible through the use of the small deep learning model that does not require any specialized hardware. The combination of visual detection with audio support further improves accessibility, and this is particularly helpful for a visually impaired person. Experimental results show that the proposed method is capable of delivering a reliable detection result under varying lighting conditions as well as with a changed background. Performance metrics prove that this approach is capable of performing effectively.

VI. Future Work

The proposed system would require future upgrades which would include improving the strength and scalability of the system. The type and amount of data used within the model could be upgraded by introducing additional currency denominations, deficient banknotes, and counterfeit banknotes. The use of advanced data augmentation methods and training the model using bigger models like YOLOv5s and YOLOv8 would be helpful.

In addition, the adoption of the system through mobile and embedded systems may add to the list of potential applications of. the system. The incorporation of multilingual text-to-speech engines and multiple currencies might make it more useful. Investigation into transformation-based vision model usage could and also a place of further research on how to improve feature. extraction.

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